

Mars

Mars is the fourth [planet](#) from the [Sun](#). It is also known as the "**Red Planet**", for [its orange-red](#) appearance.^{[22][23]} Mars is a desert-like [rocky planet](#) with a tenuous [atmosphere](#) that is primarily [carbon dioxide](#) (CO₂). At the average surface level the atmospheric pressure is a few thousandths of [Earth's](#), atmospheric temperature ranges from −153 to 20 °C (−243 to 68 °F),^[24] and [cosmic radiation](#) is high. Mars retains some water, [in the ground](#) as well as thinly in the atmosphere, forming [cirrus clouds](#), [fog](#), [frost](#), larger polar regions of [permafrost](#) and [ice caps](#) (with seasonal CO₂ snow), but no bodies of liquid surface water. Its surface gravity is roughly a third of Earth's or double that of the Moon. Its diameter, 6,779 km (4,212 mi), is about [half the Earth's](#), or twice the Moon's, and its surface area is the size of all the dry land of Earth.^[25]

[Fine dust](#) is prevalent across the surface and the atmosphere, being picked up and spread at the low Martian gravity even by the weak wind of the tenuous atmosphere. The terrain of Mars roughly follows a north–south divide, the [Martian dichotomy](#), with the northern hemisphere mainly consisting of relatively flat, low lying [plains](#), and the southern hemisphere of cratered [highlands](#). [Geologically](#), the planet is fairly active with [marsquakes](#) trembling underneath the ground, but also hosts many enormous [volcanoes](#) that are extinct (the tallest is [Olympus Mons](#), 21.9 km or 13.6 mi tall), as well as one of the largest [canyons](#) in the Solar System ([Valles Marineris](#), 4,000 km or 2,500 mi long). Mars has [two natural satellites](#) that are small and irregular in shape: [Phobos](#) and [Deimos](#). With a significant [axial tilt](#) of 25 degrees, Mars experiences [seasons](#), like Earth (which has an axial tilt of 23.5 degrees). A [Martian solar year](#) is equal to 1.88 Earth years (687 Earth days), a Martian solar day ([sol](#)) is equal to 24.6 hours.

Mars formed along with the other planets approximately 4.5 billion years ago. During the martian [Noachian](#) period (4.5 to 3.5 billion years ago), its surface was marked by [meteor impacts](#), valley formation, erosion, the [possible presence of water oceans](#) and the loss of its [magnetosphere](#). The [Hesperian](#) period (beginning 3.5 billion years ago and ending 3.3–2.9 billion years ago) was dominated by widespread [volcanic activity](#) and flooding that carved immense [outflow channels](#). The [Amazonian](#) period, which continues to the present, is the currently dominating and remaining [influence on geological processes](#). Because of Mars's geological history, the possibility of past or present [life on Mars](#) remains an area of active scientific investigation, with some possible traces needing further examination.